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Digital Museum Audiences after the Pandemic: An Analysis of Online Collection Visitor Behaviour

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ABSTRACT

During the pandemic, many museums rapidly expanded their online presence by opening access to collections, virtual exhibitions and educational programmes. Whether the enlarged digital audience has persisted since the return to in-person operations remains an open question. This article analyses anonymised web analytics logs of the online collections of five anonymised regional museums for 2019–2025, covering about 3.1 million sessions. Visitors were segmented by clustering on browsing depth, session duration, referral source and return frequency. Five segments were identified: casual visitors from search engines, students and teachers, researchers, regular enthusiasts and visitors planning an in-person visit. The total number of sessions declined after the 2020 peak but stabilised at a level 1.8 times higher than before the pandemic. Meanwhile, the share of researchers and regular enthusiasts increased, and their sessions became longer. Visitors planning a visit most often view pages with opening hours and current exhibitions, and their activity is most closely associated with subsequent museum attendance. The results show that online collections have become an independent channel of audience engagement that requires its own development strategy and performance metrics.

Keywords: museums; digital collections; web analytics; audience segmentation; digital humanities

Цифровые аудитории музеев после пандемии: анализ поведения посетителей онлайн-коллекций

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АННОТАЦИЯ

В период пандемии многие музеи резко расширили своё присутствие в интернете, открыв онлайн-доступ к коллекциям, виртуальным выставкам и образовательным программам. Вопрос о том, насколько устойчивой оказалась возросшая цифровая аудитория после возвращения к очной работе, остаётся открытым. В статье анализируются обезличенные журналы веб-аналитики онлайн-коллекций пяти условных региональных музеев за 2019–2025 гг., охватывающие около 3,1 млн сессий. Посетители сегментированы методом кластеризации по глубине просмотра, длительности сессии, источнику перехода и частоте возвращения. Выделено пять сегментов: случайные посетители из поисковых систем, учащиеся и преподаватели, исследователи, постоянные любители и посетители, планирующие очный визит. Общее число сессий после пика 2020 г. снизилось, но стабилизировалось на уровне, в 1,8 раза превышающем допандемийный. При этом доля исследователей и постоянных любителей выросла, а их сессии стали продолжительнее. Посетители, планирующие визит, чаще всего просматривают страницы с расписанием и текущими выставками, и именно их активность наиболее тесно связана с последующей посещаемостью музея. Результаты показывают, что онлайн-коллекции превратились в самостоятельный канал взаимодействия с аудиторией, требующий отдельной стратегии развития и метрик эффективности.

Ключевые слова: музеи; цифровые коллекции; веб-аналитика; сегментация аудитории; цифровые гуманитарные исследования

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1 INTRODUCTION

In spring 2020, museum closures forced institutions to move much of their public activity online. Collections databases that had previously served mainly specialists were redesigned for broader audiences, and many museums reported record numbers of website visits. As museums reopened, attention returned to physical attendance, and the fate of the digital audience became uncertain.

This study uses detailed web analytics data from five regional museums to examine how the size and composition of online collection audiences changed between 2019 and 2025, and how online behaviour relates to in-person visits.

2 MATERIALS AND METHODS

Anonymised session-level logs were obtained from five regional museums (coded M1–M5) for January 2019 to June 2025. Each session was described by the number of object pages viewed, total duration, referral source, device type and whether the visitor had returned within 90 days. Bot traffic was filtered using standard heuristics.

Sessions were clustered using k-means on standardised features; the number of clusters was chosen using silhouette scores and interpretability. Monthly in-person attendance figures were used to assess the association between online segments and subsequent visits with lagged regression models.

3 RESULTS

Annual session counts rose sharply in 2020 and then declined, stabilising from 2023 onwards at about 1.8 times the 2019 level (Fig. 1). The composition of the audience shifted: the share of researchers and regular enthusiasts grew from 14% to 23% of sessions.

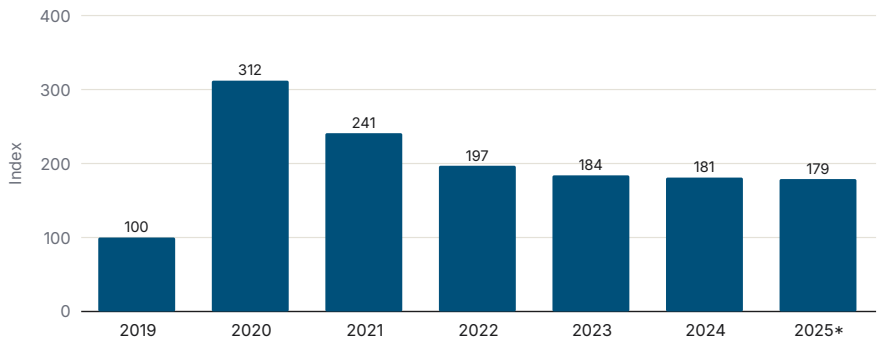


Fig. 1. Online collection sessions by year, index (2019 = 100)

Table 1 characterises the five segments. Visit planners account for only 11% of sessions, but their monthly activity was the strongest predictor of attendance in the following month.

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Table 1. Visitor segments of online collections (2023–2025)

Segment	Share, %	Median pages	Median duration
Casual (search)	46	2	1 min
Students and teachers	20	7	9 min
Researchers	9	24	31 min
Regular enthusiasts	14	12	14 min
Visit planners	11	4	3 min

4 DISCUSSION

The persistence of an enlarged digital audience suggests that the pandemic accelerated a structural change rather than causing a temporary spike. The growth of the researcher and enthusiast segments indicates that online collections increasingly serve as a destination in their own right.

For museums, this implies the need for separate metrics of digital engagement, such as depth and return frequency, rather than relying on page views or on online activity as a mere precursor of physical visits.

5 CONCLUSION

Online collections of regional museums have retained a substantially larger and more engaged audience than before the pandemic. They should be treated as an independent channel with its own goals and evaluation framework.

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